



SALES & COMMERCIAL ANALYTICS CASE STUDY

Adidas Sales Performance & Revenue Optimization

Executive Report

Simulated Role: Data Analyst — Sales / Commercial Analytics

Dataset: Public Adidas U.S. Sales Dataset (2020–2021) · Prepared with MySQL 8.0

September 2026

About this project

This is a public-dataset case study built to demonstrate a sales and commercial analytics workflow. It does not represent internal Adidas data, an actual Adidas business decision, or a real Adidas corporate outcome. All findings below are derived directly from SQL query results run against the published dataset.

REVENUE
SALES



1. Executive Summary

A. Business Problem

Sales performance for this dataset varies materially across products, retailers, regions, and sales methods. Commercial stakeholders need a structured, evidence-based view of where revenue originates, whether high-revenue segments are also the most profitable, where performance gaps exist, and which combinations warrant deeper investigation.

B. Analytical Objective

This analysis measures overall sales and profitability for the period covered by the dataset (2020–2021), identifies the key revenue contributors across product, region, retailer, and sales-method dimensions, detects high-revenue but comparatively lower-margin segments, and screens for evidence-based commercial opportunity areas that merit further, more targeted investigation.

C. Core KPIs

KPI	Value	Definition
Total Revenue	\$899.90M	SUM(total_sales)
Total Operating Profit	\$332.14M	SUM(operating_profit)
Operating Margin	36.91%	SUM(profit) / SUM(revenue) – weighted, not row-average
Reported Units Sold	2,421,548	SUM(units_sold) – source-quality caveat applies (see Section 5)
Average Price per Unit	\$45.22	AVG(price_per_unit) – descriptive only, not a causal price signal

Source: 04_kpi_analysis.sql – baseline query against the cleaned adidas_sales_clean table.

D. Key Findings

- Revenue rose from \$182.08M in 2020 to \$717.82M in 2021 (+294.2% year-over-year) with operating margin improving slightly from 34.81% to 37.44%. Only two years are present in the dataset, so this is a single year-over-year comparison, not an established multi-year trend.
- Men's Street Footwear is the largest single product, contributing \$208.83M (23.2% of total revenue) at a 39.65% margin – the highest margin among the top three revenue-generating products. Men's Athletic Footwear, the third-largest product by revenue (\$153.67M), carries the lowest product-level margin (33.74%) and is explicitly flagged by the SQL screen as 'High Revenue – Below Average Margin.'
- West is the largest revenue region (\$269.94M, 30.0% of total revenue) but has the lowest regional operating margin (33.20%) among the five regions – a scale-vs-profitability trade-off consistent with the pattern the project's SQL analysis is designed to surface.
- Commercial partners and channels show the same trade-off: West Gear is the largest retailer by revenue (\$242.96M) while Sports Direct posts the highest retailer margin (40.74%) on a smaller base; In-store leads sales-method revenue (\$356.64M) while Online carries the highest channel margin (38.99%).
- A transparent opportunity screen (above-average 2021 revenue, above-average margin, and positive year-over-year growth) flags eight product–region combinations as 'Priority for Further Investigation' – notably, all eight sit outside the West region, which otherwise dominates raw revenue.



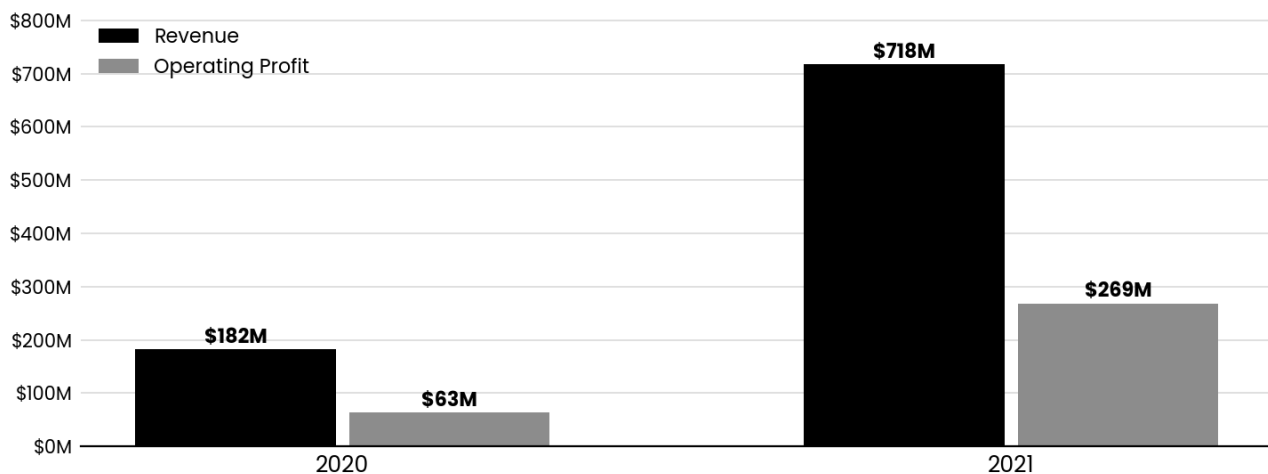
2. Revenue & Profitability

This section covers overall time performance, product performance, and regional performance — the three pillars stakeholders most often ask about first: how much did we sell, how profitable was it, and where did it come from.

A. Annual Performance (2020–2021)

Year	Revenue	Operating Profit	Operating Margin	YoY Growth
2020	\$182.08M	\$63.38M	34.81%	—
2021	\$717.82M	\$268.76M	37.44%	+294.2%

Annual Revenue vs. Operating Profit (2020–2021)



Source: 05_time_analysis.sql — annual aggregation with LAG() window function for year-over-year growth.

Finding: Revenue and operating profit both grew substantially from 2020 to 2021, and the operating margin also improved (34.81% 37.44%).

Evidence: Revenue: \$182.08M (2020) \$717.82M (2021), +294.2% YoY, calculated with LAG() over the yearly aggregate.

Business implication: The 2021 increase is large enough that it should not be treated as a stable growth rate going forward — with only two years of data, it is one year-over-year comparison, not a multi-year trend line. The margin improvement alongside the revenue increase suggests growth was not purely a low-margin volume expansion.

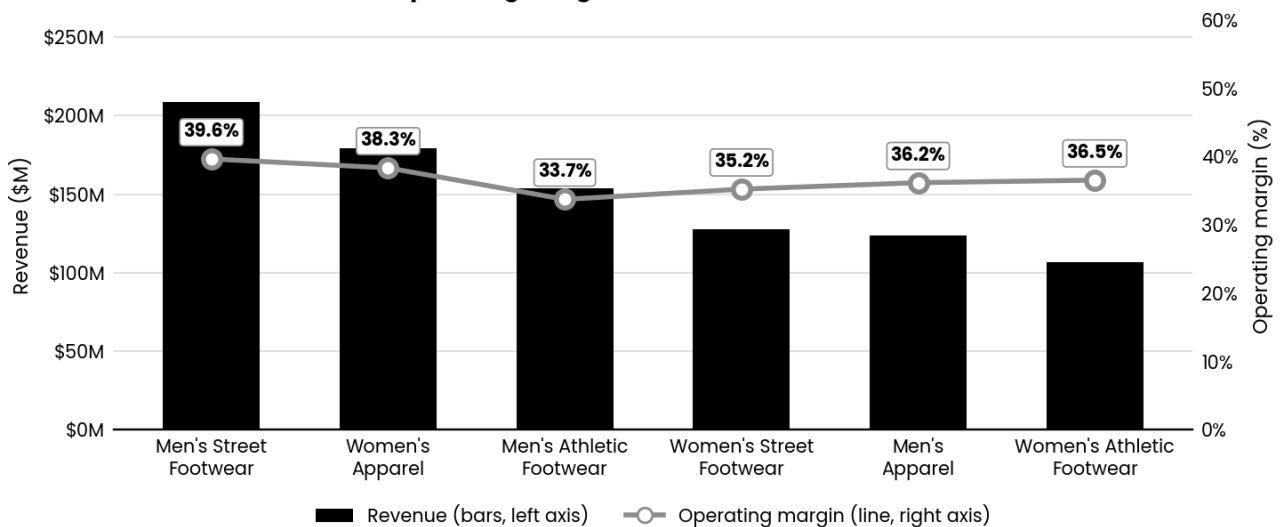
Recommended next action: Investigate which regions, products, and channels contributed most to the 2021 increase (see subsections B–D below and Section 3) before assuming the growth rate will repeat in future periods.



B. Product Performance

Product	Revenue	Op. Profit	Margin	Rev. Contribution
Men's Street Footwear	\$208.83M	\$82.80M	39.65%	23.2%
Women's Apparel	\$179.04M	\$68.65M	38.34%	19.9%
Men's Athletic Footwear	\$153.67M	\$51.85M	33.74%	17.1%
Women's Street Footwear	\$128.00M	\$45.10M	35.23%	14.2%
Men's Apparel	\$123.73M	\$44.76M	36.18%	13.7%
Women's Athletic Footwear	\$106.63M	\$38.98M	36.55%	11.8%

Product Revenue and Operating Margin



Source: 06_product_analysis.sql — product scorecard with revenue_contribution window function; revenue_rank and profit_rank were also computed and matched exactly for all six products (no product's revenue rank diverges from its profit rank).

Finding: Revenue and profit rankings are aligned across all six products. Men's Athletic Footwear is the exception on margin: it ranks third by revenue but has a 33.74% margin and is flagged High Revenue / Below Average Margin.

Evidence: Men's Street Footwear: \$208.83M revenue, 39.65% margin, 23.2% contribution. Men's Athletic Footwear: \$153.67M revenue, 33.74% margin — flagged in 11_opportunity_analysis.sql's high-revenue/below-average-margin screen. Revenue rank and profit rank are identical (1–6) across all six products.

Business implication: Because revenue and profit ranks align for every product, scale and profitability are broadly consistent at the product level. Men's Athletic Footwear is the exception on margin, not on rank: its margin trails its revenue scale.

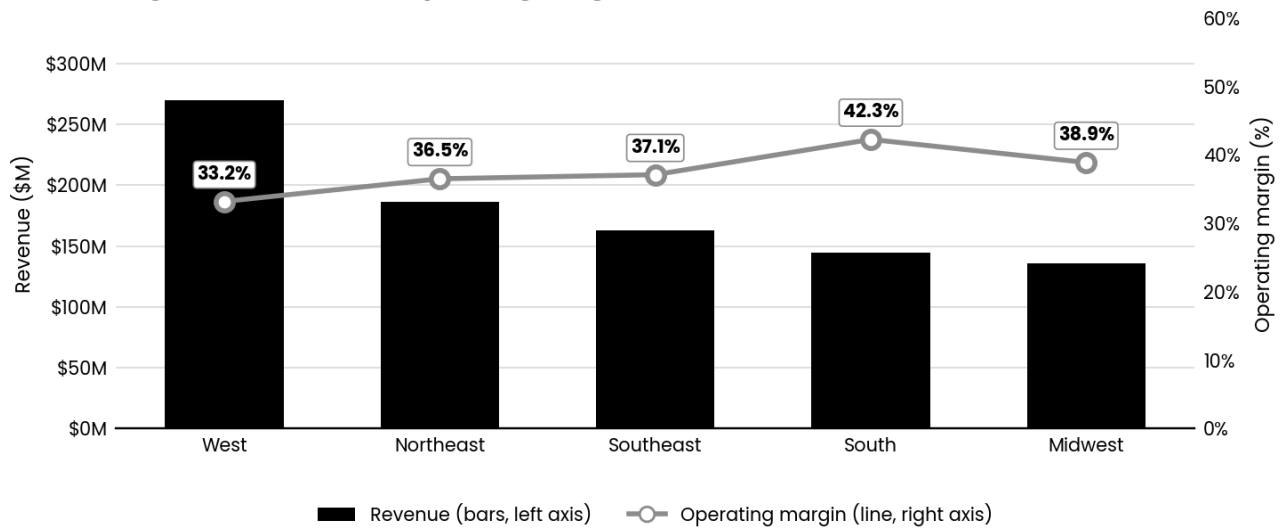
Recommended next action: Drill into Men's Athletic Footwear by region, retailer, and sales method to identify where its margin gap is concentrated before considering any commercial or pricing conversation.



C. Regional Performance

Region	Revenue	Op. Profit	Margin	Rev. Contribution
West	\$269.94M	\$89.61M	33.20%	30.0%
Northeast	\$186.32M	\$68.02M	36.51%	20.7%
Southeast	\$163.17M	\$60.56M	37.11%	18.1%
South	\$144.66M	\$61.14M	42.26%	16.1%
Midwest	\$135.80M	\$52.81M	38.89%	15.1%

Regional Revenue and Operating Margin



Source: 07_region_analysis.sql – regional scorecard with revenue_contribution window function.

Top 5 states by revenue (state-level drill-down):

State	Region	Revenue	Op. Profit	Margin
New York	Northeast	\$64.23M	\$23.33M	36.32%
California	West	\$60.17M	\$19.30M	32.08%
Florida	Southeast	\$59.28M	\$20.93M	35.30%
Texas	South	\$46.36M	\$18.69M	40.31%
South Carolina	Southeast	\$29.29M	\$11.32M	38.67%

Source: 07_region_analysis.sql – state × region aggregation, LIMIT 15 (top 5 shown here).

Finding: West is the largest revenue region but has the lowest aggregate operating margin among the five regions.

Evidence: West: \$269.94M revenue (30.0% of total), 33.20% margin – the lowest of the five regions, versus a high of 42.26% in the South.

Business implication: Because West is the largest region in absolute terms, even a modest margin gap represents a meaningful amount of profit relative to other regions. West is not being called a 'bad' region – its revenue scale still matters commercially – but the margin gap deserves explanation.

Recommended next action: Drill into West by state, product, retailer, and sales method (the 07_region_analysis.sql state-level and product × region diagnostics support this) to identify whether the margin gap is concentrated in specific combinations rather than spread evenly.

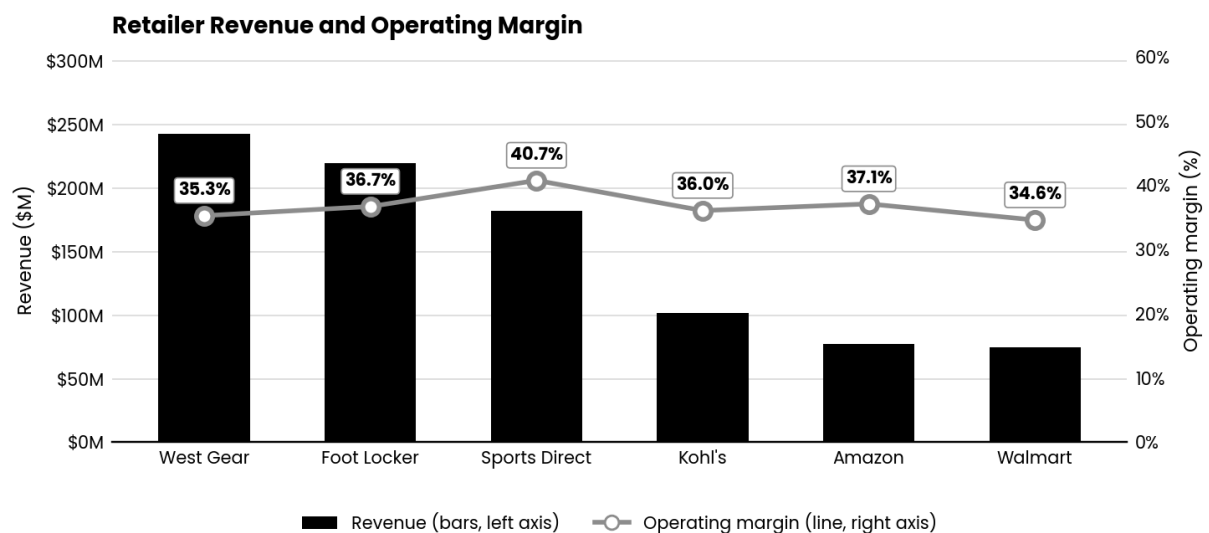


3. Commercial Drivers

This section examines the two partner-facing drivers that most directly shape commercial conversations: retailers and sales methods (channels).

A. Retailer Performance

Retailer	Revenue	Op. Profit	Margin	Rev. Contribution
West Gear	\$242.96M	\$85.67M	35.26%	27.0%
Foot Locker	\$220.09M	\$80.72M	36.68%	24.5%
Sports Direct	\$182.47M	\$74.33M	40.74%	20.3%
Kohl's	\$102.11M	\$36.81M	36.05%	11.3%
Amazon	\$77.70M	\$28.82M	37.09%	8.6%
Walmart	\$74.56M	\$25.78M	34.58%	8.3%



Source: 08_retailer_analysis.sql — retailer scorecard with revenue_contribution window function.

A retailer × sales-method breakdown (08_retailer_analysis.sql) shows that mix varies by partner — for example, Sports Direct's largest channel is Outlet (44.24% margin), while Amazon's largest channel is Online (40.55% margin) — indicating that a retailer's overall margin is influenced by which channels it relies on most.

Finding: West Gear is the largest retailer by revenue, while Sports Direct carries the highest retailer-level margin on a smaller revenue base.

Evidence: West Gear: \$242.96M revenue (27.0% contribution), 35.26% margin. Sports Direct: \$182.47M revenue (20.3% contribution), 40.74% margin — the highest among the six retailers.

Business implication: Revenue scale and profitability are not the same measure of commercial performance. A large partner is not automatically the most profitable one, and a high-margin partner is not automatically the biggest revenue source.

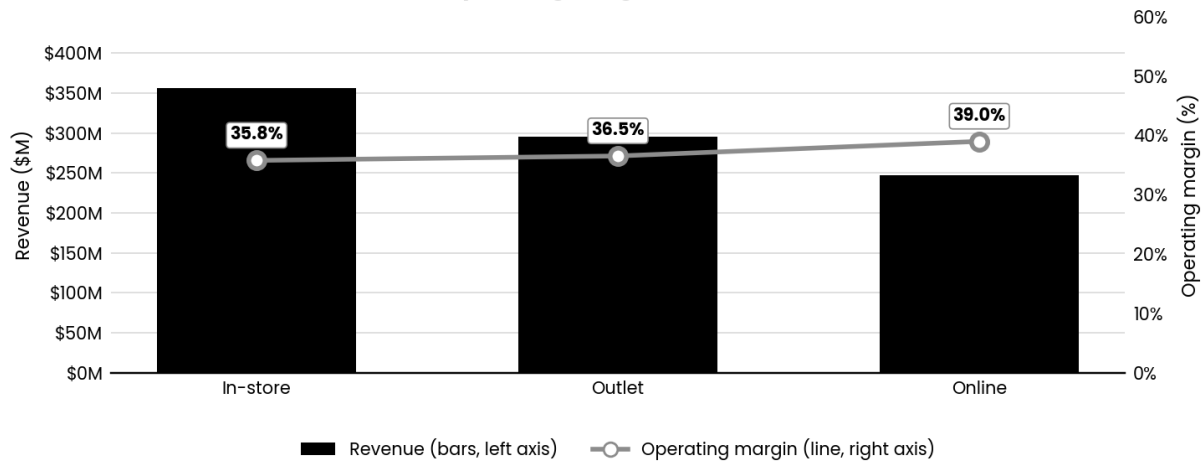
Recommended next action: Compare retailer × product × sales-method mix (not just the headline retailer number) before any discussion of commercial terms, discounting, or resource allocation across retail partners.



B. Sales Method / Channel Performance

Sales Method	Revenue	Op. Profit	Margin	Reported Units
In-store	\$356.64M	\$127.59M	35.78%	672,767
Outlet	\$295.59M	\$107.99M	36.53%	821,341
Online	\$247.67M	\$96.56M	38.99%	927,440

Sales Method: Revenue and Operating Margin



Source: 09_sales_method_analysis.sql — channel scorecard. Reported Units should be read with the source-quality caveat in Section 5.

Year-over-year growth by channel (2021 vs. 2020):

Sales Method	2021 Revenue	2020 Revenue	YoY Growth
Online	\$243.15M	\$4.52M	+5,279.5%
In-store	\$262.74M	\$93.90M	+179.8%
Outlet	\$211.93M	\$83.66M	+153.3%

Source: 09_sales_method_analysis.sql — channel × year aggregation with LAG().

Finding: In-store generates the largest sales-method revenue, while Online carries the highest aggregate margin and, on paper, the fastest year-over-year growth rate.

Evidence: In-store: \$356.64M revenue, 35.78% margin. Online: \$247.67M revenue, 38.99% margin, and 2021 YoY growth of +5,279.5% — driven by a very small 2020 base of \$4.52M growing to \$243.15M in 2021.

Business implication: Channel decisions should weigh scale and profitability together, not revenue alone. The Online growth percentage is mathematically correct but is inflated by a small starting base, so it should not be read as evidence that Online is now the dominant channel in absolute terms.

Recommended next action: Investigate product and retailer composition within each sales method to explain the margin difference, and treat the Online growth rate as a base-effect finding rather than a forecast of future channel share.



4. Opportunity Areas — Create an Opportunity Screen

The opportunity screen implemented in `11_opportunity_analysis.sql` is a transparent screening rule, not a mathematical 'best segment' score. It flags 2021 product–region combinations that simultaneously show above-average revenue and above-average margin (relative to the average of all product–region combinations in that year) together with positive year-over-year growth ($\text{growth} > 0$). A flag means a combination deserves a closer look — it is not proof that the segment caused strong performance or that further investment will reproduce the same result.

A. Product-Level Screen: High Revenue, Below-Average Margin

At the product level, the same screening logic (Section 12.3 / Appendix A8 of the project guide) flags Men's Athletic Footwear as 'High Revenue – Below Average Margin' — its revenue (\$153.67M) sits above the average product revenue, while its margin (33.74%) sits below the overall portfolio margin. None of the other five products meet both conditions simultaneously.

B. Product × Region Opportunity Screen (2021)

Product	Region	2021 Revenue	Margin	YoY Growth	Screen Result
Men's Street Footwear	Northeast	\$42.58M	40.29%	+404.0%	Priority
Men's Street Footwear	Midwest	\$35.82M	39.35%	+1,333.7%	Priority
Women's Apparel	Northeast	\$29.53M	38.79%	+268.3%	Priority
Men's Street Footwear	Southeast	\$29.08M	39.53%	+319.0%	Priority
Women's Apparel	Midwest	\$26.73M	43.61%	+1,710.9%	Priority
Women's Apparel	Southeast	\$25.20M	42.51%	+300.8%	Priority
Women's Apparel	South	\$24.64M	49.31%	+395.5%	Priority
Men's Street Footwear	South	\$24.21M	41.34%	+471.3%	Priority

Source: `11_opportunity_analysis.sql` — product × region 2021 screen (above-average revenue AND above-average margin AND positive YoY growth). All eight flagged combinations sit outside the West region. No West product–region combinations were flagged by the screen. The flag requires above-average 2021 revenue, above-average margin, and positive YoY growth simultaneously; the underlying West combinations should therefore be reviewed against these criteria.

Finding: Eight product–region combinations for 2021 meet all three opportunity-screen criteria, and all eight are outside the West region — concentrated instead in the Northeast, Midwest, Southeast, and South.

Evidence: Examples: Men's Street Footwear × Northeast (\$42.58M revenue, 40.29% margin, +404.0% YoY); Women's Apparel × South (\$24.64M revenue, 49.31% margin, +395.5% YoY) — the highest margin among all flagged combinations.

Business implication: Despite West's dominant raw revenue, none of its product–region combinations met all three screen criteria simultaneously in 2021, while several smaller regional combinations did. This is diagnostic evidence for where to look next, not proof that these combinations are the 'best' segments or that the pattern will repeat in a future period.

Recommended next action: Treat the eight flagged combinations as a shortlist for further investigation — validate the underlying state, retailer, and sales-method mix behind each one (as demonstrated in `06_product_analysis.sql`'s product × region diagnostic) before drawing any commercial conclusion or reallocating resources. Separately, review the West combinations against the three criteria to see which condition each one did not meet.



5. Data Quality & Limitations

The findings above should be read together with the following data-quality considerations, all identified during the validation and cleaning stage of this project (02_data_validation.sql, 03_data_cleaning.sql, 10_price_volume_quality.sql).

Raw file and population

- The raw uploaded file contained 9,652 rows. Four rows were completely blank across every field and were excluded, leaving an analytical population of 9,648 records — this is the row count used throughout the clean table and confirmed in 02_data_validation.sql.
- The raw file's helper column ('Unnamed: 13') was entirely empty and is not used anywhere in this analysis.
- Numeric financial fields (price, total sales, operating profit, operating margin) were stored as text with mixed punctuation in the raw file and required explicit parsing before any aggregation could be trusted.

Margin validation

- The supplied Operating Margin field was validated against $\text{Operating Profit} \div \text{Total Sales}$; the two agree closely, which supports using Total Sales, Operating Profit, and Operating Margin as the core, internally coherent commercial metrics for this report.
- As an additional consistency check, Total Revenue reconciles exactly to \$899.90M whether it is summed by year, by product, by region, by retailer, or by sales method — a useful confirmation that the clean table aggregates correctly across every dimension used in this report.

Units Sold anomaly

- Units Sold does not sit on a single consistent scale in the source file (some records report small counts such as 1–2 units, others report several hundred). Because of this, $\text{Price per Unit} \times \text{Units Sold}$ does not reliably reconstruct Total Sales.
- Running that arithmetic consistency check directly against the cleaned table (10_price_volume_quality.sql) confirms the issue at scale: 3,886 of the 9,648 records (about 40.3%) show a difference greater than \$1 between Total Sales and Price $\times$ Units Sold.
- Reported Units Sold and Average Price per Unit are therefore presented in this report as descriptive, source-quality-flagged fields only. They are not used to reconstruct revenue, and no volume-productivity or price-elasticity claim is made from them.

Scope and interpretation limits

- The dataset covers only 2020 and 2021. Every year-over-year growth figure in this report is a single two-point comparison, not an established multi-year trend, and should be read accordingly — particularly the Online channel's 2021 growth rate, which is amplified by a very small 2020 base.
- This is a public case-study dataset. Findings describe patterns within this dataset only. They are not evidence of real Adidas corporate strategy, internal decisions, or actual business outcomes, and no causal claim is made about why any product, region, retailer, or channel performed as it did.
- The opportunity screen in Section 4 is a transparent screening rule based on average-revenue and average-margin thresholds plus a positive-growth condition within the 2021 data — it is not a predictive model and does not guarantee that a flagged segment will outperform in a future period.

Closing Note

Together, these limitations do not undermine the core commercial findings — revenue, operating profit, and operating margin are internally consistent and well validated across every cut used in this report. They do mean that Units Sold, price-based volume claims, and any single-year growth percentage should be treated as descriptive and directional rather than definitive, and that every recommendation in this report is deliberately framed as an area for further investigation rather than a final business decision.